1 (20 points) TRUE/FALSE

State if the following statements are true or false. Please write TRUE or FALSE in the space provided. Each problem is worth 1 point, but you will be penalized -1 point for an incorrect answer, so leave the answer blank if you don't know.

(a) A good MapReduce programmer is recommended to write a GroupByKey function as well as Map and Reduce.

Answer:_____

(b) The MapReduce framework explicitly checks whether the Reduce function is commutative and associative before applying a Combiner.

Answer:_____

(c) Spark outperforms MapReduce even when the intermediate data is too large to fit in memory.

Answer:_____

(d) In Spark, suppose that there is a `sort` operation that shuffles the order of all elements in several RDDs. Does it have a wide dependency?

Answer:_____

(e) All maximal frequent itemsets are closed.

Answer:_____

(f) A-priori aims to find all *interesting* rules.

Answer:_____

(g) In the multistage algorithm, we can further reduce false positives by increasing the number of hash tables when finding frequent itemsets.

Answer:_____

(h) In Toivonen's algorithm, we need to check if the frequent itemsets in the sample are truely frequent in the full data to avoid false negatives.

Answer:_____

(i) The character-level k-shingles are necessary for running LSH for documents.

Answer:_____

(j) We can't represent shingles as 4-byte tokens if the number of unique shingles exceeds 2^{32}.

Answer:_____

(k) In LSH, we can estimate the shape of an S-curve with the value $1 - (1 - s^r)^b$.

Answer:_____

(l) The random hyperplane technique can also be used directly for Jaccard similarity.

Answer:_____

(m) The hierarchical clustering is not effective for an Euclidean space.

Answer:_____

(n) k-means++ is always more effective than k-means.

Answer:_____

(o) The CURE algorithm can still work if we don't move the points 20% inward.

Answer:_____

(p) The GRGPF algorithm can be directly used for the other distance measures.

Answer:_____

(q) A data matrix of size $m \times n$, containing m independent data points, can have up to n latent factors.

Answer:_____

(r) Running PCA with the power iteration if often faster than running it with SVD.

Answer:_____

(s) SVD can be applied to any real matrix.

Answer:_____

(t) For dimensionality reduction, CUR decomposition can be more accurate than SVD in terms of the reconstruction error in some cases.

Answer:_____

2 [20 points] MapReduce and Spark

(a) [5 points] We want to perform a linear transformation of a vector $\mathbf{v} \in \mathbb{R}^6$ using a matrix $\mathbf{A} \in \mathbb{R}^{3\times 6}$. That is, we want to compute $\mathbf{Av}$.

$$\mathbf{v} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 7 & 8 & 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 & 17 & 18 \end{bmatrix}.$$

Suppose that we have a laptop with restricted main memory, which has the following constraints: (a) $\mathbf{A}$ and $\mathbf{v}$ are stored in its disk. (b) It can load from disk and store only 6 numbers at a time. (c) It can perform any arithmetic operation on the stored numbers, but has to write it into disk if it already has 6 numbers. Our idea is to use MapReduce to design a disk-based algorithm to compute $\mathbf{Av}$.

Please describe the MapReduce process, including the use of a **combiner**, and solve the matrix multiplication to obtain the correct result.

(b) [7 points] In MapReduce, when a Map worker fails, all the tasks assigned to that worker need to be re-executed. On the other hand, when a Reduce worker fails, only the in-progress tasks need to be re-executed. Explain why this difference exists.

(c) [8 points] The KAIST High School recently conducted a math competition for all its students. The scores of students from grades 1, 2, and 3 are stored in a text file; each line contains `(Student ID, Grade, Score)`. Using MapReduce programming, write pseudocode to calculate the variance of scores for each grade.

Note: Your pseudocode doesn't have to be perfect in terms of grammar, but the immediate results of functions used in the pseudocode must be clear.

(i) Map function:
- Input of the Map function should be `(Student ID, Grade, Score)`.

(ii) Reduce function:
- Output of the Reduce function should be `(Grade, Variance of Grade)`.

3 [20 points] Frequent Itemsets

(a) [10 points] Given the following transaction database, we aim to find frequent itemsets with the support threshold $s = 3$ using the Toivonen's algorithm.

Transaction ID	**Items**
T1	{A, B, D}
T2	{A, C, D, E}
T3	{B, C, D}
T4	{A, B, E}
T5	{A, B, C}
T6	{B, D, E}
T7	{A, D, E}
T8	{B, C, E}

(i) Suppose that we have a sample consisting of T1, T2, T4 and T5 (i.e., 50% of the data). Using a support threshold of $s' = 1$ (i.e., less than 50% of s), find all frequent itemsets in the sample and construct the negative border.

(ii) Based on the negative border constructed in (i), prove that there are no false negatives in the frequent itemsemts found in (i).

(iii) Toivonen's algorithm is not guaranteed to finish in certain conditions. Think of two new transactions to be added to the dataset, T9 and T10, which make the algorithm not terminated assuming that we have the same sample (i.e., T1, T2, T4 and T5) and $s' = 1$ as in (i).

Note: T9 and T10 should be different, each containing at most 4 items.

(b) [10 points] Assume the same set of transactions and the support threshold $s = 3$ as in Problem (a). Do not consider T9 and T10 you created for (a-iii).

(i) Identify all **strong association rules** from the frequent itemsets in the data, where a strong rule is defined as one with confidence of at least 70%. For each rule, calculate the confidence as a percentage rounded to two decimal places.

(ii) Determine all the closed frequent itemsets in the dataset with support at least $s = 3$. Provide your answer as a list of closed frequent itemsets.

4 [25 points] Finding Similar Items

(a) [5 points] Suppose that you use locality-sensitive hashing (LSH) with **cosine similarity** using random hyperplanes for 100-dimensional vectors to find near-duplicate documents based on their Jaccard similarity. You have two sets, A and B, with their respective minhash signatures computed using 100 hash functions.

Derive the relationship between the number of hash functions k needed to ensure that two vectors with cosine similarity s are placed into the same bucket with a probability of at least 0.9. Express your answer symbolically, and you do not need to compute the exact numerical result.

(b) [8 points] The precision and recall of the LSH scheme can be defined as follows:

- **Precision:** The proportion of correctly identified similar items out of all the items identified as similar by the LSH scheme.
- **Recall:** The proportion of correctly identified similar items out of all actual similar items available in the dataset.

(i) Consider the **AND-OR construction** for the LSH family, where you vary the number of rows per band r and bands b. Assume that each row in each band uses a unique, independent hash function, leading to a total of $k = r \times b$ independent hash functions across the scheme.

Explain how increasing the number of rows per band r while keeping the total number of hash functions $k = r \times b$ fixed affects the **precision** and **recall**. Discuss the impact on the **collision probability** of similar and dissimilar items, and how these changes affect the rates of **false positives** and **false negatives**.

(ii) Now consider the case of the **OR-AND construction**. Explain how increasing r affects the precision and recall in this scenario, focusing on the same aspects: collision probability, false positives, and false negatives.

(c) [12 points] Consider the following distance metric for vectors $x, y \in \mathbb{R}^d$:

$$D_\delta(x, y) = \sum_{i=1}^{d} \mathbb{I}(|x_i - y_i| > \delta) \tag{1}$$

where $\mathbb{I}(\cdot)$ is an indicator function that returns 1 if the condition inside is true and 0 otherwise, and δ is a non-negative threshold. This distance counts the dimensions where the difference between the corresponding components of x and y exceeds δ.

(i) A function $D(x, y)$ can be considered a valid distance metric if it satisfies the following properties:

- **Non-negativity:** $D(x, y) \geq 0$
- **Identity of indiscernible:** $D(x, y) = 0$ iff $x = y$
- **Symmetry:** $D_\delta(x, y) = D(y, x)$
- **Triangle inequality:** $D(x, z) \leq D(x, y) + D(y, z)$

For each property, determine whether $D_\delta(x, y)$ satisfies the condition when $\delta > 0$. If it does, provide a **proof**. If not, provide a specific **counterexample**.

(ii) Consider a family of hash functions $\mathcal{F} = \{f_i\}$, defined by the following function:

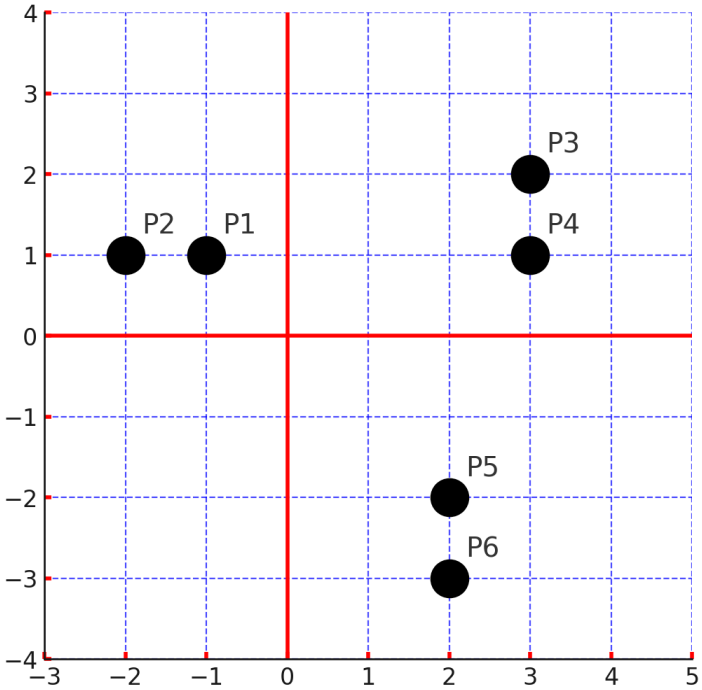
$$f_i(x) = \mathbb{I}(x_{k_i} \geq \tau_i) \tag{2}$$

where each τ_i is a threshold chosen randomly from the possible values of the k_i-th dimension. For each hash function f_i, k_i is randomly chosen between 1 to d, and it evaluates whether the component x_{k_i} exceeds the threshold τ_i.

Suppose we apply this family $\mathcal{F}$ to vectors x and y, and the distance $D_0(x, y)$ is defined as $D_\delta(x, y)$ with $\delta = 0$. This family $\mathcal{F}$ is considered (d_1, d_2, p_1, p_2)-sensitive. **Express** p_1 and p_2 in terms of d_1, d_2, and d (number of dimensions). Provide a clear justification and explanation of your reasoning for full credit.

5 [20 points] Clustering

Suppose you are given the following six points: P1$(-1, 1)$ to P6$(2, -3)$. You will simulate `k-means++` and the BFR algorithm to identify clusters in the data.



(a) [10 points] The results of the standard `k-means` algorithm heavily depend on the choice of initial centroids. In the following problems, we will explore the `k-means++` algorithm to address this issue by selecting better initial centroids.

(i) [4 points] Assume **P2** is chosen as the first initial centroid. Calculate the **probability** of each remaining point being selected as the next initial centroid based on the Euclidean distance, and identify the second initial centroid $\boldsymbol{P_C}$.

$P1$: _____ $P3$: _____ $P4$: _____

$P5$: _____ $P6$: _____ P_C: _____

(ii) [6 points] Starting with **P2** and $\boldsymbol{P_C}$ from (i), perform `k-means++` using the Euclidean distance. You must show intermediate steps to get full credit.

(b) [10 points] The BFR algorithm is particularly effective for large datasets. It maintains a compact representation of clusters to minimize memory usage.

(i) [4 points] Classify the points into the **Discard Set**, **Compressed Set**, and **Retained Set**. The initial cluster centroids are **P2** and $\boldsymbol{P_C}$ as in Problem (a-i). If the Euclidean distance to a centroid is less than **2**, the point is considered close enough. After classification, find the **N, SUM**, and **SUMQ** of each clusters or miniclusters. DO NOT perform any merging at this stage.

(ii) [6 points] Using the **Discard Set** and **Compressed Set** from part (b-i), determine whether any clusters or miniclusters should be merged based on the condition below. If no merges should be made, state this explicitly.

Condition: If the Euclidean distance between their centroids, D_E, satisfies the following, perform the merge:

$$D_E < 4\left(\sigma_i + \sigma_j\right),$$

where σ_i and σ_j are the standard deviations of two clusters C_i and C_j, respectively. Here, each cluster's standard deviation σ_i is defined as:

$$\sigma_i = \sqrt{\sigma_{i,x}^2 + \sigma_{i,y}^2}$$

where $\sigma_{i,x}$ and $\sigma_{i,y}$ are the standard deviations of cluster C_i along the x and y axes, respectively. YOU MUST use **N, SUM**, and **SUMQ** from (b-i).

6 [20 points] Dimensionality Reduction

(a) [10 points] Suppose we have a 4×3 matrix M, which represents preference scores for 4 students (in rows) on 3 different subjects (in columns). We decompose the matrix M using **SVD decomposition**, and the result is as below.

$$M = U\Sigma V^T = \begin{bmatrix} 0.5 & 0.2 & 0.2 \\ 0.6 & 0.0 & 0.2 \\ 0.5 & 0.8 & 0.1 \\ 0.4 & 0.4 & 0.9 \end{bmatrix} \begin{bmatrix} 4x & 0 & 0 \\ 0 & 2x & 0 \\ 0 & 0 & x \end{bmatrix} \begin{bmatrix} 0.4 & 0.8 & 0.1 \\ 0.7 & 0.1 & 0.6 \\ 0.1 & 0.0 & 0.4 \end{bmatrix} \tag{3}$$

(i) [5 points] Suppose that we do not know the value of x ($x \geq 0$) in Σ. Though, we know that when we project each student's preference vector on the **concept axes**, the minimum L1 distance between any pair of the 4 students becomes 4.0. **What percentage of total energy** do we retain if we eliminate the last (lowest) singular value? Derive x and percentage of total energy, and describe your answer. (L1 distance: $d_1(p, q) = ||p - q||_1 = \sum_{i=1}^{n} |p_i - q_i|$)

x: _____

$percentage(\%)$: _____

Description:

(ii) [5 points] After eliminating the last (lowest) singular value in (i), we can think of it as leaving two *concepts*. Suppose that 4 new students (P1 $\sim$ P4) come to school and assign their preference scores on the subjects as:

$$S_1 = \begin{bmatrix} 2 & 1 & 3 \end{bmatrix}, \quad S_2 = \begin{bmatrix} 1 & 2 & 4 \end{bmatrix}, \quad S_3 = \begin{bmatrix} 2 & 5 & 2 \end{bmatrix}, \quad S_4 = \begin{bmatrix} 1 & 3 & 2 \end{bmatrix} \tag{4}$$

How can we reasonably divide these 4 students into two groups if we only use the remaining two concepts? Explain your answer in relation to the *concept*, not just the similarity of their preference scores.

Group 1: _____, Group 2: _____

Description:

(b) [10 points] Suppose we have a 5×4 matrix A below. Let us compute the **CUR decomposition** of A when we pick the columns for C_2 and C_3 and the rows for R_3 and R_5 (in this order). Write down your answer and description on each question.

$$A = \begin{array}{c} \\ R_1 \\ R_2 \\ R_3 \\ R_4 \\ R_5 \end{array} \begin{array}{c} \begin{array}{cccc} C_1 & C_2 & C_3 & C_4 \end{array} \\ \begin{bmatrix} 1 & 1 & 2 & 2 \\ 2 & 2 & 0 & 0 \\ 1 & 3 & 4 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 4 & -3 & 1 \end{bmatrix} \end{array}$$

(i) [2.5 points] Derive C which has a size of 5×2, and describe your answer.

C: _____

Description:

(ii) [2.5 points] Derive R which has a size of 2×4, and describe your answer.

R: _____

Description:

(iii) [5 points] Derive U and describe your answer. (For simplicity, you may use *power iteration* with $x_0 = \begin{bmatrix} 1 & 1 \end{bmatrix}^T$ and the number of iterations of 2 (i.e., until x_2).)

U: _____

Description:

[Scratch page]